

Unit 5 Practice Problems

Similarity Practice

How to use this file: Copy the notes carefully, then cover the worked examples and redo them without looking. For practice files, work first, then check the answer bank. Detailed solutions are included for Unit 3 through Unit 8 practice as requested.

Unit 5 Practice Problems

Directions

Work all problems first. Then check the answer bank. Detailed solutions are included after the answer bank.

Problems

1. Dilate $(-2, 5)$ by scale factor 3 centered at the origin.
2. Dilate $(8, -4)$ by scale factor $\frac{1}{2}$ centered at the origin.
3. Dilate $P(6, 9)$ by scale factor $\frac{1}{2}$ centered at $C(2, 1)$.
4. A segment of length 12 is dilated by scale factor 2.5. Find the image length.
5. If a figure is dilated by length scale factor 4, what is the area scale factor?
6. Triangles have corresponding side lengths 6, 9, 12 and 8, 12, 16. Find the scale factor from the first triangle to the second.
7. Similar triangles have corresponding sides 5 and 8, and another pair x and 24. Find x .
8. Two triangles have angles $40^\circ, 70^\circ, 70^\circ$ and $40^\circ, 70^\circ, 70^\circ$. Why are they similar?
9. Two triangles have sides 6 and 10 around an included angle. Another triangle has corresponding sides 9 and 15 around a congruent included angle. Which similarity theorem applies?
10. Are triangles with side lengths 4, 6, 8 and 10, 15, 20 similar? If yes, give the scale factor from first to second.
11. Are triangles with side lengths 4, 6, 9 and 8, 12, 15 similar?
12. In $\triangle ABC$, $DE \parallel BC$. If $AD = 6$, $DB = 4$, $AE = 9$, find EC .
13. If a midsegment of a triangle is 14, what is the length of the parallel third side?
14. In a right triangle, the altitude to the hypotenuse divides the hypotenuse into segments 9 and 16. Find the altitude and both legs.
15. Similar polygons have side ratio 3 : 5. If the smaller perimeter is 42, find the larger perimeter.
16. Similar figures have area ratio 9 : 25. Find the length ratio.
17. Similar figures have area ratio 16 : 49. If the smaller side is 12, find the corresponding larger side.
18. What happens to a line not passing through the center under a dilation?
19. What happens to a line passing through the center under a dilation?
20. Triangle $(0, 0), (2, 0), (0, 3)$ maps to $(1, 1), (5, 1), (1, 7)$. Describe a sequence of transformations.
21. If $AB = 5$ and its image $A'B' = 15$, find the scale factor.
22. Which theorem proves similarity when two side ratios around an included angle are equal and the included angles are congruent?
23. Why are the two smaller triangles formed by the altitude to the hypotenuse of a right triangle similar to the original triangle?

- 24.** A 6-ft person casts a 4-ft shadow. A tree casts a 30-ft shadow. Find the tree height.
- 25.** Dilate $P(4, 0)$ by scale factor 3 centered at $(1, -2)$.

Answer Bank

1. $(-6, 15)$.
2. $(4, -2)$.
3. $(4, 5)$.
4. 30.
5. 16.
6. $\frac{4}{3}$.
7. 15.
8. AA similarity.
9. SAS similarity.
10. Yes; scale factor $\frac{5}{2}$.
11. No.
12. 6.
13. 28.
14. Altitude 12; legs 15 and 20.
15. 70.
16. $3 : 5$.
17. 21.
18. It maps to a parallel line.
19. It remains the same line.
20. Dilate by 2 about the origin, then translate by $\langle 1, 1 \rangle$.
21. 3.
22. SAS similarity.
23. They share acute angles, so AA similarity applies.
24. 45 ft.
25. $(10, 4)$.

Detailed Solutions

1. A dilation centered at the origin multiplies both coordinates by 3: $(-2, 5) \mapsto (-6, 15)$.
2. Multiply both coordinates by $\frac{1}{2}$: $(8, -4) \mapsto (4, -2)$.
3. Use $(a + k(x - a), b + k(y - b))$ with center $(2, 1)$ and $k = \frac{1}{2}$. The image is $(2 + \frac{1}{2}(6 - 2), 1 + \frac{1}{2}(9 - 1)) = (4, 5)$.
4. Lengths multiply by the scale factor: $12(2.5) = 30$.
5. Area scales by the square of the length scale factor, so $4^2 = 16$.
6. Compare corresponding sides: $8/6 = 12/9 = 16/12 = 4/3$. The scale factor from first to second is $\frac{4}{3}$.
7. Set up corresponding ratios: $\frac{5}{8} = \frac{x}{24}$. Cross multiply: $8x = 120$, so $x = 15$.
8. Two pairs of corresponding angles are congruent, so the triangles are similar by AA. In fact, all three angle measures match.
9. The side ratios are $6/9 = 10/15 = 2/3$, and the included angles are congruent. This is SAS similarity.
10. Check ratios: $10/4 = 15/6 = 20/8 = 2.5$. All three corresponding side ratios match, so the triangles are similar by SSS similarity with scale factor $\frac{5}{2}$.
11. The ratios $8/4 = 2$, $12/6 = 2$, and $15/9 = 5/3$ do not all match. The triangles are not similar.
12. Since $DE \parallel BC$, the sides are divided proportionally: $\frac{AD}{DB} = \frac{AE}{EC}$. Thus $\frac{6}{4} = \frac{9}{x}$, so $6x = 36$ and $x = 6$.
13. A triangle midsegment is half the length of the parallel third side. If the midsegment is 14, the third side is 28.
14. Let the hypotenuse segments be $p = 9$ and $q = 16$, so $c = 25$. The altitude satisfies $h^2 = pq = 144$, so $h = 12$. The legs satisfy $a^2 = cp = 25(9) = 225$, so $a = 15$, and $b^2 = cq = 25(16) = 400$, so $b = 20$.
15. Perimeters scale like lengths. If smaller:larger is $3 : 5$, then $42/P = 3/5$. So $3P = 210$, and $P = 70$.
16. Area ratio is the square of length ratio. Since $9 : 25 = 3^2 : 5^2$, the length ratio is $3 : 5$.
17. Area ratio $16 : 49$ gives length ratio $4 : 7$. If 12 corresponds to the smaller side, then $12/L = 4/7$, so $4L = 84$, and $L = 21$.
18. A dilation maps a line not passing through the center to a parallel line. The distance from the center changes, but the direction remains parallel.
19. A line through the center maps onto itself because every point moves along the same line through the center.
20. First dilate the original triangle by scale factor 2 about the origin: $(0, 0), (2, 0), (0, 3) \mapsto (0, 0), (4, 0), (0, 6)$. Then translate by $\langle 1, 1 \rangle$, giving $(1, 1), (5, 1), (1, 7)$.
21. Scale factor is image length divided by original length: $15/5 = 3$.
22. This is SAS similarity because the included angles are congruent and the two side pairs around the included angles are proportional.
23. The altitude creates right triangles. Each smaller triangle shares one acute angle with the original triangle, and both have a right angle, so AA similarity applies.
24. Use similar triangles: $\frac{6}{4} = \frac{h}{30}$. Cross multiply: $4h = 180$, so $h = 45$ ft.

- 25.** Use center formula with $(a, b) = (1, -2)$, $P(4, 0)$, and $k = 3$: $(1 + 3(4 - 1), -2 + 3(0 + 2)) = (10, 4)$.